

VENKATA PAVAN KANDULA

DevOps Engineer II | SRE | Kubernetes & Cloud Infrastructure Specialist

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PROFESSIONAL SUMMARY

Results-driven DevOps Engineer with 4+ years of hands-on experience scaling and securing enterprise Kubernetes infrastructure on AWS EKS, implementing GitOps workflows with ArgoCD, and managing high-throughput datastores (Elasticsearch, RDS, DocumentDB) and messaging systems (RabbitMQ, Kafka). Proven track record at TCS managing 100+ microservices across multi-environment promotion chains, with deep expertise in Terraform IaC, GitHub Actions CI/CD, observability stacks (Datadog, Grafana, ELK), Kubernetes autoscaling (HPA/VPA), disaster recovery planning, and security compliance frameworks. Delivered a 40% reduction in release cycle time, 30% AWS cost savings through right-sizing and Reserved Instance optimization, and eliminated 8–10 hours of weekly manual toil through Python/Bash automation.

CORE TECHNICAL SKILLS

Kubernetes / EKS: EKS, GKE, AKS cluster design, RBAC, network policies (Calico/Flannel), Istio service mesh, Helm, pod security standards, ArgoCD GitOps, HPA/VPA autoscaling, cluster autoscaler

CI/CD: GitHub Actions (primary), GitLab CI/CD, Jenkins, AWS CodePipeline, ArgoCD — multi-env pipelines
Dev → QA → UAT → Prod

Cloud (AWS): EC2, EKS, S3, RDS, Aurora, DocumentDB, Lambda, API Gateway, IAM, VPC, CloudWatch, Secrets Manager, AWS Cost Explorer, Trusted Advisor, Reserved Instances

Infrastructure as Code: Terraform (modules, workspaces, remote state), Helm chart authoring, Ansible, AWS CloudFormation, GitOps workflows

Disaster Recovery & Backup: RDS automated backups, Aurora point-in-time recovery, multi-AZ failover, DR runbooks, RTO/RPO planning, cross-region S3 replication

Security & Compliance: OPA/Gatekeeper policy enforcement, zero-trust architecture, container image scanning (Trivy), TLS/mTLS, RBAC, IAM least-privilege, secrets management, SOC 2 controls alignment

Cost Optimization: AWS cost governance, right-sizing, Reserved Instance planning, Spot Instance strategy, resource tagging, Cost Explorer dashboards

Datastores: Elasticsearch (ELK), AWS RDS, Aurora (Multi-AZ, read replicas), DocumentDB, PostgreSQL, Databricks Lakehouse

Messaging Systems: RabbitMQ, Kafka — queue management, throughput tuning, dead-letter queue configuration

Monitoring & Observability: Datadog, Grafana, Prometheus, ELK Stack, CloudWatch, Splunk — real-time dashboards & alerting

Scripting & Automation: Python (self-healing systems, predictive scaling, automation), Bash/Shell, Groovy (Jenkins shared libraries)

Networking & Security: VPC design, ALB/NLB, DNS, TLS/mTLS, OPA/Gatekeeper, zero-trust architecture, container image scanning

PROFESSIONAL EXPERIENCE

Senior DevOps / Site Reliability Engineer | Tata Consultancy Services (TCS) — Hyderabad, India | Mar 2022 – Present (4 Years)

- Kubernetes Autoscaling (HPA/VPA):** Designed and deployed Horizontal Pod Autoscaler (HPA) and Vertical Pod Autoscaler (VPA) policies across 100+ microservices on AWS EKS; combined with Cluster Autoscaler for node-level scaling — ensuring SLA adherence during traffic spikes and reducing over-provisioning by ~25%.
- Kubernetes Cluster Management (EKS/GKE/AKS):** Provisioned and managed production-grade Kubernetes clusters on AWS EKS; enforced RBAC, network policies, CNI plugins (Calico/Flannel), Istio service mesh integration, and pod security standards; performed cluster upgrades with zero downtime.
- ArgoCD GitOps Deployments:** Implemented and maintained ArgoCD-based GitOps workflows for consistent, automated application deployments across Dev → QA → UAT → Prod; ensured single source of truth for all Kubernetes manifests and Helm charts, reducing config drift and rollback time.
- AWS Cost Optimization (30% Savings):** Led cloud cost optimization initiative using AWS Cost Explorer and Trusted Advisor — identified and right-sized over-provisioned EC2 and RDS instances, implemented Reserved Instance purchasing strategy, and introduced Spot Instances for non-critical workloads; achieved ~30% reduction in monthly AWS spend.

- Disaster Recovery & Backup Strategy: Architected and implemented DR strategy across environments — configured RDS automated backups with point-in-time recovery, Aurora Multi-AZ failover, and cross-region S3 replication for critical artifacts; defined RTO/RPO targets and authored DR runbooks for P1 incident scenarios.
- Security Compliance & Hardening: Enforced security compliance across Kubernetes clusters using OPA/Gatekeeper policy-as-code for admission control; implemented container image scanning with Trivy in CI pipelines; applied IAM least-privilege principles, rotated secrets via AWS Secrets Manager, and aligned infrastructure controls with SOC 2 requirements.
- Datastore Management (Elasticsearch, RDS, DocumentDB): Administered Elasticsearch clusters for ELK observability stack; managed AWS RDS (PostgreSQL, Aurora Multi-AZ with read replicas) and DocumentDB; performed performance tuning, backup policies, and scaling operations to support high-throughput workloads.
- Messaging Systems (RabbitMQ, Kafka): Configured and managed RabbitMQ and Kafka queuing systems for event-driven microservices architectures; tuned throughput settings, monitored queue depths, implemented dead-letter queue strategies, and ensured high availability for mission-critical pipelines.
- Terraform Infrastructure as Code: Authored modular Terraform configurations for all AWS infrastructure (VPC, EKS, RDS, S3, IAM, Lambda, API Gateway); used workspaces and remote state for environment isolation; maintained GitOps-aligned IaC repositories enabling repeatable, auditable provisioning.
- GitHub Actions CI/CD Pipelines: Designed and maintained end-to-end CI/CD pipelines using GitHub Actions, GitLab CI/CD, and Jenkins; integrated automated build, test, security scanning (Trivy, SAST), and deployment stages — achieving a 40% reduction in release cycle time; on-boarded multiple applications onto the DevOps toolchain.
- Monitoring & Observability (Datadog / Grafana / ELK): Deployed enterprise observability stack using Datadog, Prometheus + Grafana, and ELK Stack; built real-time dashboards for cluster health, pipeline metrics, and datastore performance; implemented intelligent alerting that significantly reduced MTTR.
- Python & Bash Automation: Developed Python and Bash scripts eliminating 8–10 hours of weekly manual toil for infrastructure health checks, artifact management, and operational reporting; built self-healing Kubernetes operators for automatic workload recovery and incident remediation.
- Engineering Team Support & On-Call: Provided 24/7 on-call support; resolved P1/P2 incidents within defined SLA; conducted RCA for critical failures; mentored junior engineers on Kubernetes, Terraform, and CI/CD best practices; produced architecture diagrams and operational runbooks.

KEY ACHIEVEMENTS

- 40% reduction in release cycle time through end-to-end CI/CD pipeline optimization across GitHub Actions, GitLab CI/CD, and Jenkins.
- ~30% AWS cost savings achieved via right-sizing, Reserved Instance strategy, and Spot Instance adoption — identified through Cost Explorer and Trusted Advisor analysis.
- Eliminated 8–10 hours of weekly manual toil via Python/Bash automation for infrastructure health checks, artifact management, and reporting.
- Kubernetes autoscaling (HPA/VPA + Cluster Autoscaler) deployed across 100+ microservices, reducing over-provisioning by ~25% while maintaining SLA compliance.
- Disaster recovery strategy implemented with defined RTO/RPO targets — Aurora Multi-AZ failover, RDS point-in-time recovery, and cross-region S3 replication for mission-critical data.
- Security compliance hardening using OPA/Gatekeeper, Trivy image scanning, IAM least-privilege, and SOC 2-aligned controls across production Kubernetes infrastructure.
- Zero-downtime application on-boarding — successfully on-boarded multiple applications onto ArgoCD GitOps with 100% error-free, zero-production-incident record.
- Enterprise observability implemented using Datadog + Prometheus + Grafana + ELK Stack, driving significant MTTR reduction across production Kubernetes infrastructure.
- Recognized at TCS for driving CI/CD transformation and Kubernetes infrastructure automation initiatives organization-wide.

EDUCATION

Bachelor of Technology (B.Tech) — Mechanical Engineering

Aditya Engineering College, JNTU Kakinada | Graduated October 2020